Database Administration: Server Backups, User Roles and Permissions

Detailed Hands-On Learning Guide for Corporate SQL Training

Database Administration (DBA) is responsible for managing, securing, monitoring, and maintaining databases in enterprise applications. A Database Administrator ensures: - Data safety - Backup and recovery - Security and permissions - High availability - Performance optimization

Roles and Responsibilities of DBA

• Database installation and configuration

• Backup and recovery management

• Performance tuning

• Security management

• Managing user roles and permissions

• Database monitoring

• Disaster recovery planning

1. Database Server Backups

A backup is a copy of database data stored safely to recover data during: - Server crashes - Hardware failures - Cyber attacks - Human errors - Data corruption

Types of Database Backups

Backup Type	Description
Full Backup	Complete database copy
Incremental Backup	Backs up only changed data
Differential Backup	Backs up changes since last full backup
Transaction Log Backup	Backs up transaction logs

MySQL Full Backup Example

```
mysqldump -u root -p ecommerce_db > backup.sql
```

This command creates a complete backup of the ecommerce_db database.

Restore Backup Example

```
mysql -u root -p ecommerce_db < backup.sql
```

Used to restore database from backup file.

Real-Life Backup Use Cases

• Banking systems

• E-commerce websites

• Hospital management systems

• Airline reservation systems

• ERP and finance applications

2. User Roles and Permissions

User roles and permissions control who can access and modify database data. Security is critical in enterprise systems to prevent: - Unauthorized access - Data theft - Accidental data deletion

Create User Example

```
CREATE USER 'trainer'@'localhost'
IDENTIFIED BY 'password123';
```

Grant Permissions Example

```
GRANT SELECT, INSERT, UPDATE
ON ecommerce_db.*
TO 'trainer'@'localhost';
```

This user can now read, insert, and update records.

Revoke Permissions Example

```
REVOKE INSERT, UPDATE
ON ecommerce_db.*
FROM 'trainer'@'localhost';
```

Common Permission Types

Permission	Purpose
SELECT	Read data
INSERT	Add records
UPDATE	Modify records
DELETE	Remove records
ALL PRIVILEGES	Full access

Hands-On Practice Exercises

• Create a database backup using mysqldump

• Restore database from backup file

• Create new database users

• Grant SELECT permission to user

• Grant INSERT and UPDATE permissions

• Revoke DELETE permission

Mini Project - Secure Banking Database

• Create Banking database

• Create Accounts and Transactions tables

• Take full backup

• Restore backup into another database

- Create admin and employee users
- Assign different permissions
- Test database security

Common Mistakes

- Not taking regular backups
- Granting ALL PRIVILEGES unnecessarily
- Using weak passwords
- Not testing backup recovery
- Ignoring database monitoring

Performance and Security Tips

- Schedule automated backups
- Store backups in secure locations
- Use strong passwords
- Grant minimum required permissions
- Monitor suspicious login activity
- Regularly test backup recovery

Interview Questions

- What is database backup?
- Difference between full and incremental backup?
- What is mysqldump?
- What is GRANT command?
- Difference between GRANT and REVOKE?
- Why are user permissions important?
- What are database security best practices?

Final Summary

Topic	**Purpose**
Server Backup	Protect data and enable recovery
Restore	Recover database from backup
User Roles	Control database access
Permissions	Secure database operations